conception of space: for him, space provided the language for specifying the location of one object relative to another but was not itself an independent entity. Initially, Einstein was an enthusiastic champion of Mach's perspective, because it was the most relative that a theory espousing relativity could be. But as Einstein's understanding of general relativity deepened, he realized that it did not incorporate Mach's ideas fully. According to general relativity, the water in Newton's bucket, spinning in an otherwise empty universe, would take on a concave shape, and this conflicts with Mach's purely relational perspective, since it implies an absolute notion of acceleration. Even so, general relativity does incorporate some aspects of Mach's viewpoint, and within the next few years a more than $500 million experiment that has been in development for close to forty years will test one of the most prominent Machian features.

The physics to be studied has been known since 1918, when the Austrian researchers Joseph Lense and Hans Thirring used general relativity to show that just as a massive object warps space and time—like a bowling ball resting on a trampoline—so a rotating object drags space (and time) around it, like a spinning stone immersed in a bucket of syrup. This is known as *frame dragging* and implies, for example, that an asteroid freely falling toward a rapidly rotating neutron star or black hole will get caught up in a whirlpool of spinning space and be whipped around as it journeys downward. The effect is called frame dragging because from the point of view of the asteroid—from its frame of reference—it isn't being whipped around at all. Instead, it's falling straight down along the spatial grid, but because space is swirling (as in Figure 14.1) the grid gets twisted, so the meaning of "straight down" differs from what you'd expect based on a distant, nonswirling perspective.

To see the connection to Mach, think about a version of frame dragging in which the massive rotating object is a huge, hollow sphere. Calculations initiated in 1912 by Einstein (even before he completed general relativity), which were significantly extended in 1965 by Dieter Brill and Jeffrey Cohen, and finally completed in 1985 by the German physicists Herbert Pfister and K. Braun, showed that space inside the hollow sphere would be dragged by the rotational motion and set into a whirlpool-like spin.[1] If a stationary bucket filled with water—stationary